EXPERIMENT 11: Monthly Sales Visualization Using Matplotlib

1. Line Plot of Monthly Sales

Aim

To write a Python program to create a line plot showing monthly product sales using Matplotlib.

Algorithm

1. Import the Matplotlib library.
2. Get month names and sales values from the user.
3. Create a line plot using plt.plot().
4. Add a title and axis labels.
5. Display the plot using plt.show().

Program

Writing

```
import matplotlib.pyplot as plt


months = input("Enter month names: ").split()

sales = list(map(int, input("Enter sales values: ").split()))


plt.plot(months, sales, marker='o')

plt.title("Monthly Sales - Line Plot")

plt.xlabel("Months")

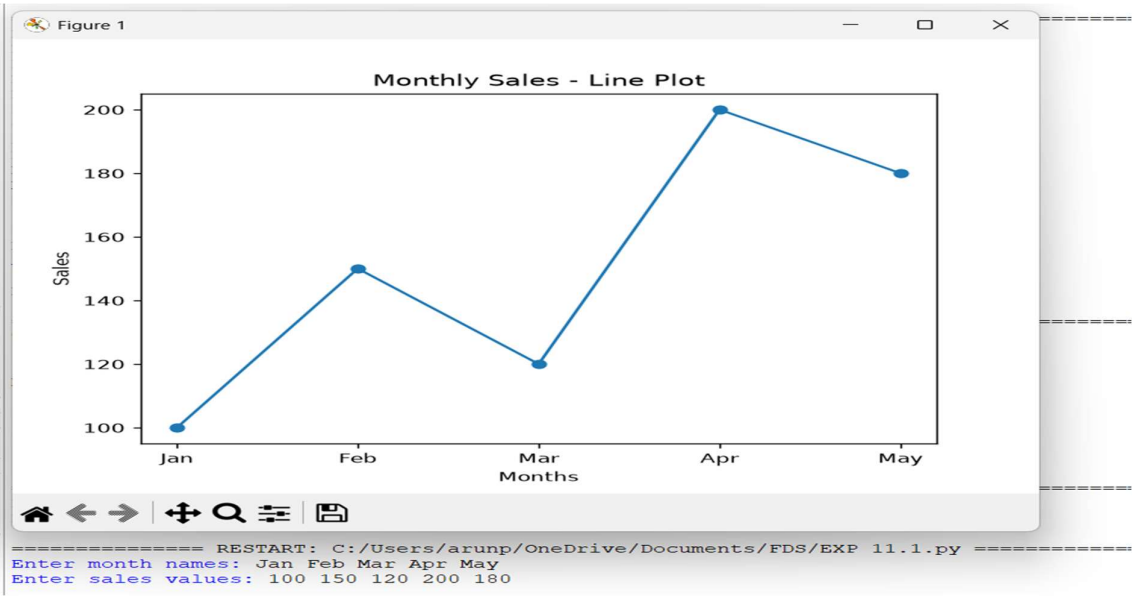
plt.ylabel("Sales")

plt.show()
```

Sample Input

Enter month names: Jan Feb Mar Apr May

Enter sales values: 100 150 120 200 180



Result

The line plot displaying monthly product sales is generated successfully.

2. Scatter Plot of Monthly Sales

Aim

To write a Python program to create a scatter plot showing monthly product sales using Matplotlib.

Algorithm

1. Import Matplotlib.
2. Get month names and sales values.
3. Create a scatter plot using plt.scatter().
4. Add a title and axis labels.
5. Display the plot.

Program

Writing

```
import matplotlib.pyplot as plt


months = input("Enter month names: ").split()
```

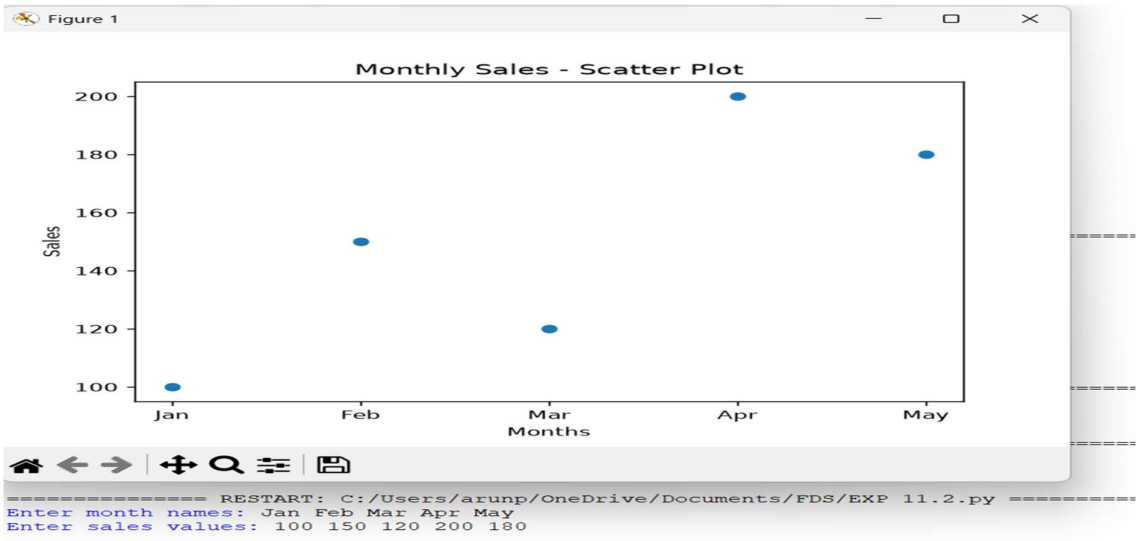
```
sales = list(map(int, input("Enter sales values: ").split()))


plt.scatter(months, sales)
plt.title("Monthly Sales - Scatter Plot")
plt.xlabel("Months")
plt.ylabel("Sales")
plt.show()
```

Sample Input

Enter month names: Jan Feb Mar Apr May

Enter sales values: 100 150 120 200 180



Result

The scatter plot displaying the monthly sales values is generated successfully.

3. Bar Plot of Monthly Sales

Aim

To write a Python program to create a bar plot representing monthly product sales using Matplotlib.

Algorithm

1. Import Matplotlib.

2. Get month names and sales values.
3. Create a bar plot using plt.bar().
4. Add a title and axis labels.
5. Display the plot.

Program

Writing

```
import matplotlib.pyplot as plt


months = input("Enter month names: ").split()

sales = list(map(int, input("Enter sales values: ").split()))


plt.bar(months, sales)

plt.title("Monthly Sales - Bar Plot")

plt.xlabel("Months")

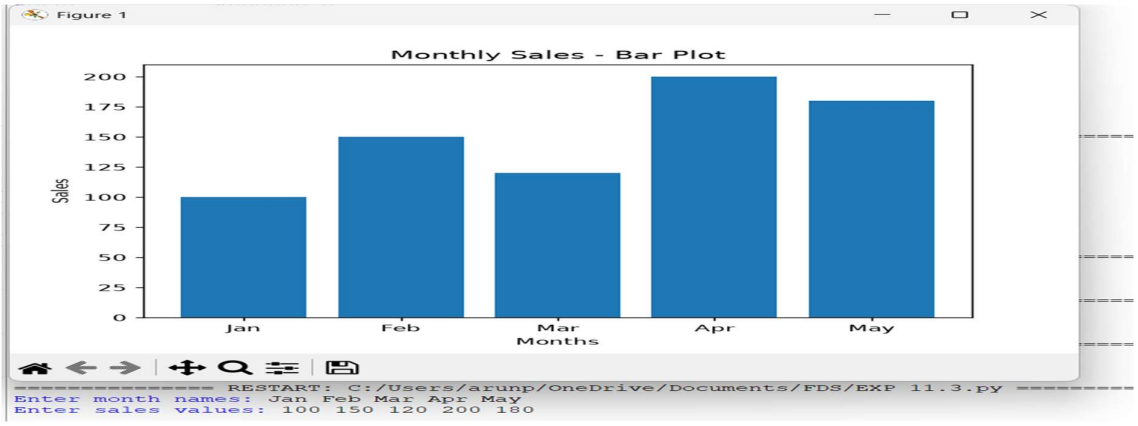
plt.ylabel("Sales")

plt.show()
```

Sample Input

Enter month names: Jan Feb Mar Apr May

Enter sales values: 100 150 120 200 180



Result

The bar plot representing monthly product sales is generated successfully.